

Disturbances in 3D sensing

Reproducible test environment for environmental disturbances
caused by Lighting Effect



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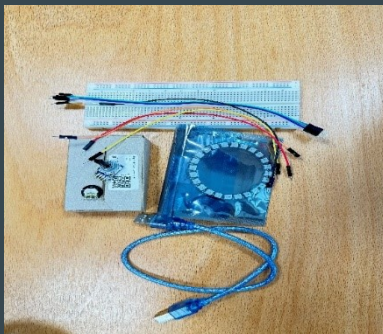
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Problems

- **Wavelength issue:** Inconsistent wavelength leads to inaccurate surface detection.
 - **Blinking effect:** Causes unstable lighting during 3D scanning.
- 3D scanner struggles to capture precise geometry and texture.

Lighting Source



Items for lighting set up

1. Setup

Items acquired:

- NeoPixel Ring (24 LEDs)
- Breadboard & jumper wires
- Cable & Arduino Uno

With help from Mr. Nils Suhr, a stable stand for the ring light was constructed.

Ring light connected with jumper wires.

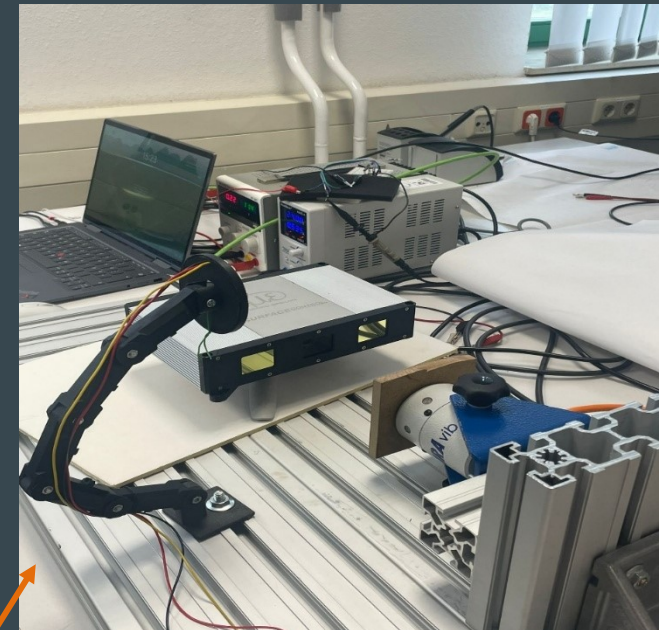


3D Printer

Lighting Source

2. Programming

- Connected Arduino Uno to laptop via cable.
- Uploaded Arduino code to control the NeoPixel ring light.
- Configured brightness and patterns to simulate different lighting conditions.
- This became the controlled light source for the 3D scan experiment.



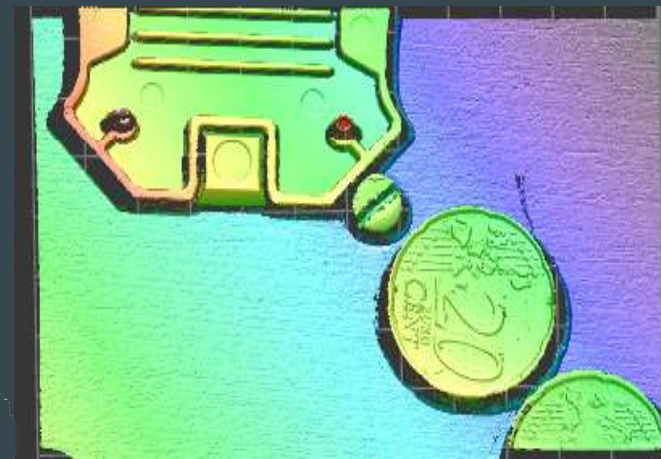
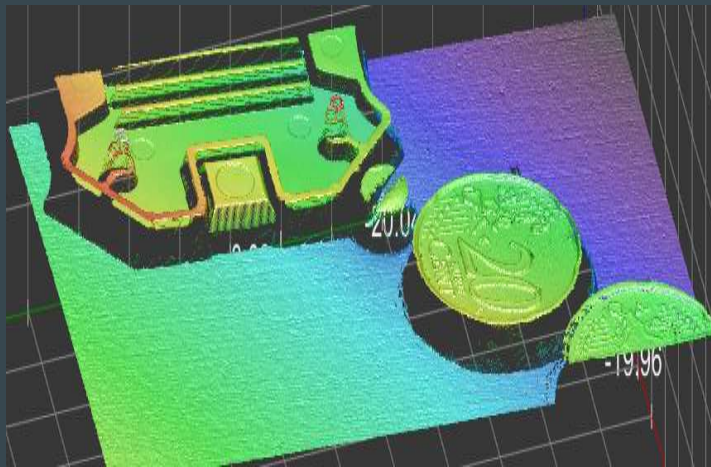
Set up for the experiment

Light source for the experiment

Observation

Exposure: 2700 μ s

Sensor Light Brightness: 10 (for all cases)



LED light / No Blinking / No Room light

Observation

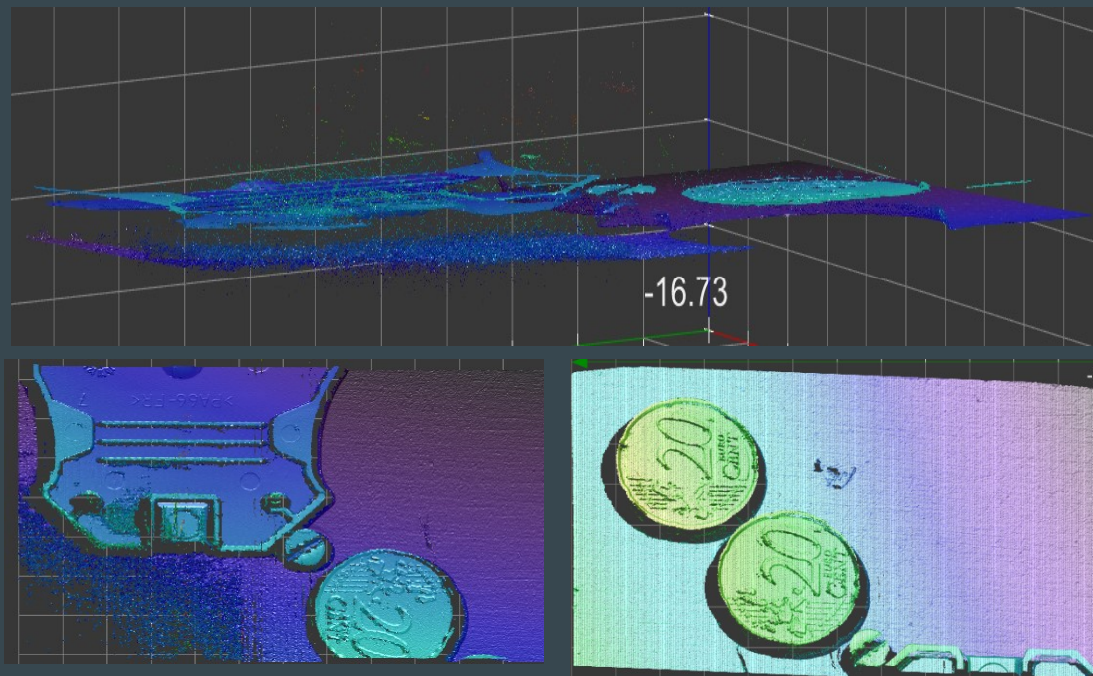
Exposure: 2700 μ s

Sensor Light Brightness: 10 (for all cases)



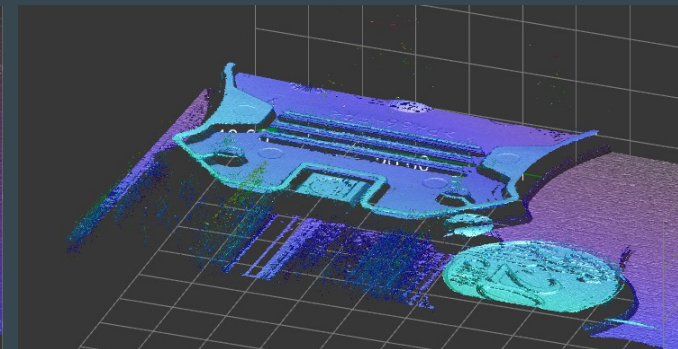
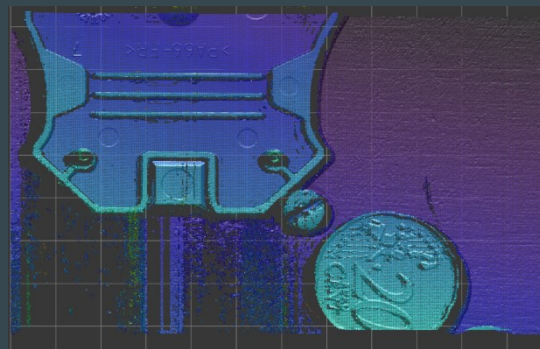
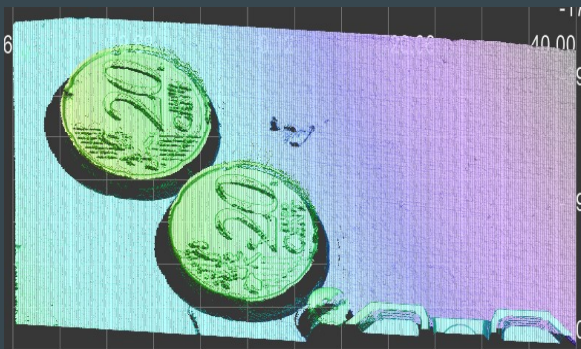
Blue light / Blinking 20ms (Scan 7)

Observation



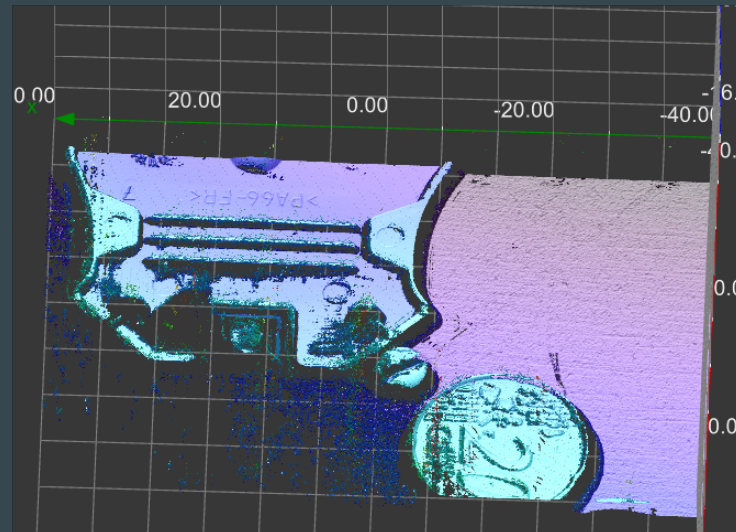
Blue light / 30ms Blinking (Scan 6)

Observation



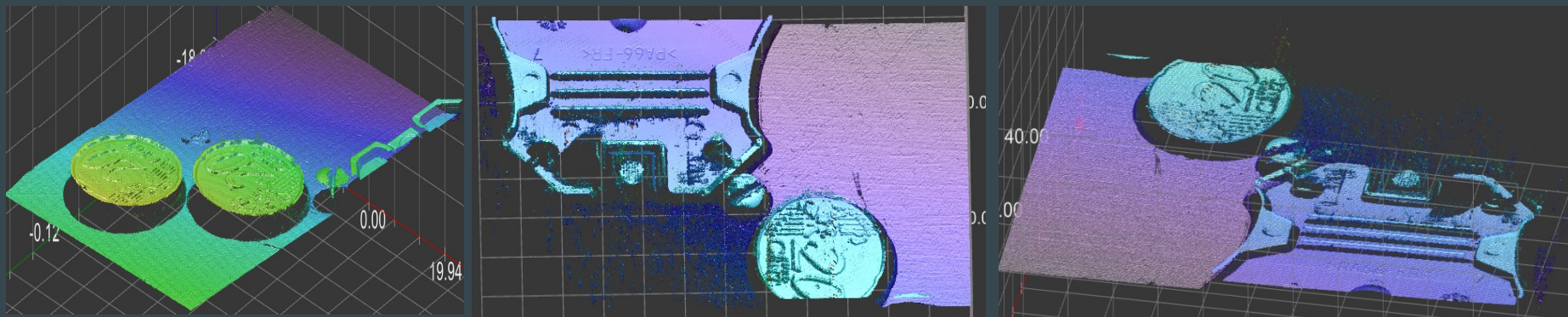
Blue Light / 50ms Blinking / Full brightness 255 (Scan 3)

Observation



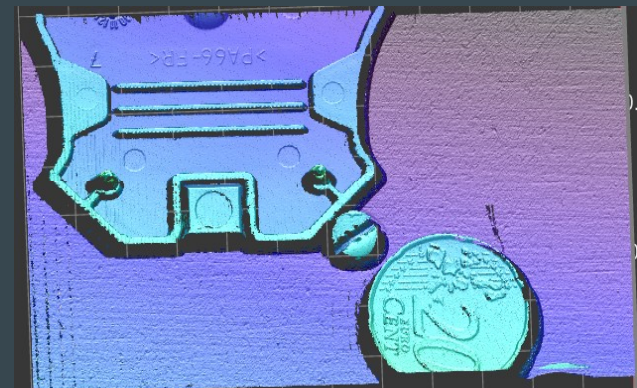
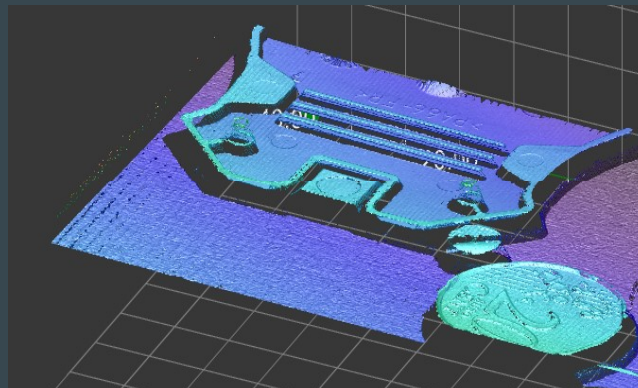
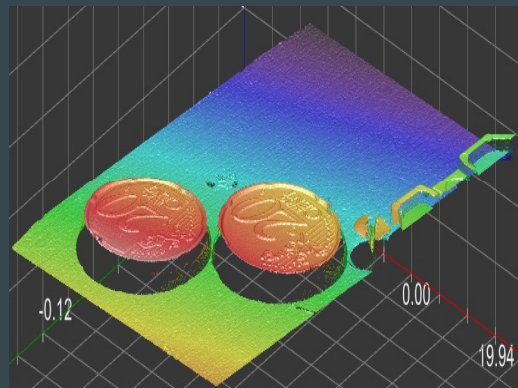
Blue light / 60ms Blinking / No room light

Observation



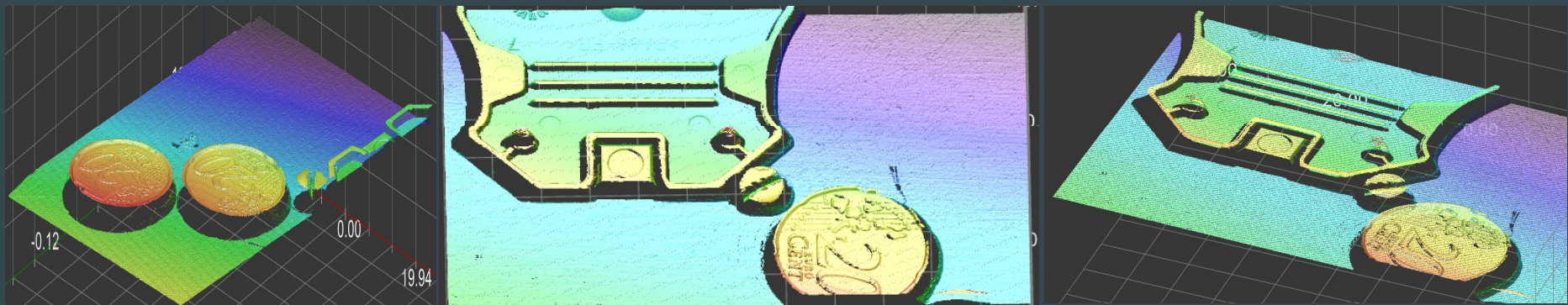
Blue light / 100ms Blinking (Scan 8)

Observation



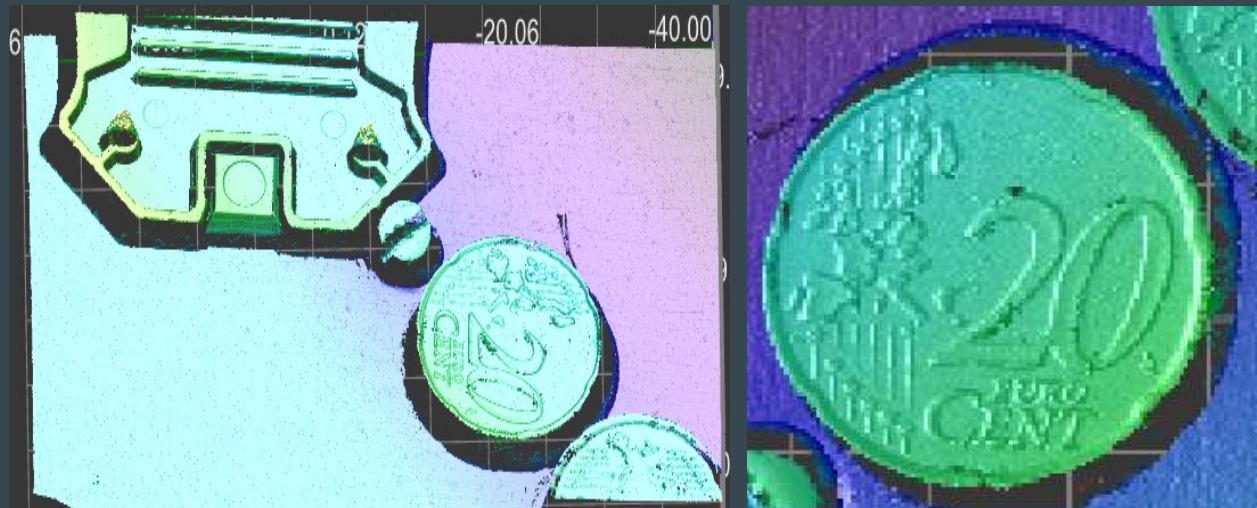
Blue light / 500ms (Scan 9)

Observation



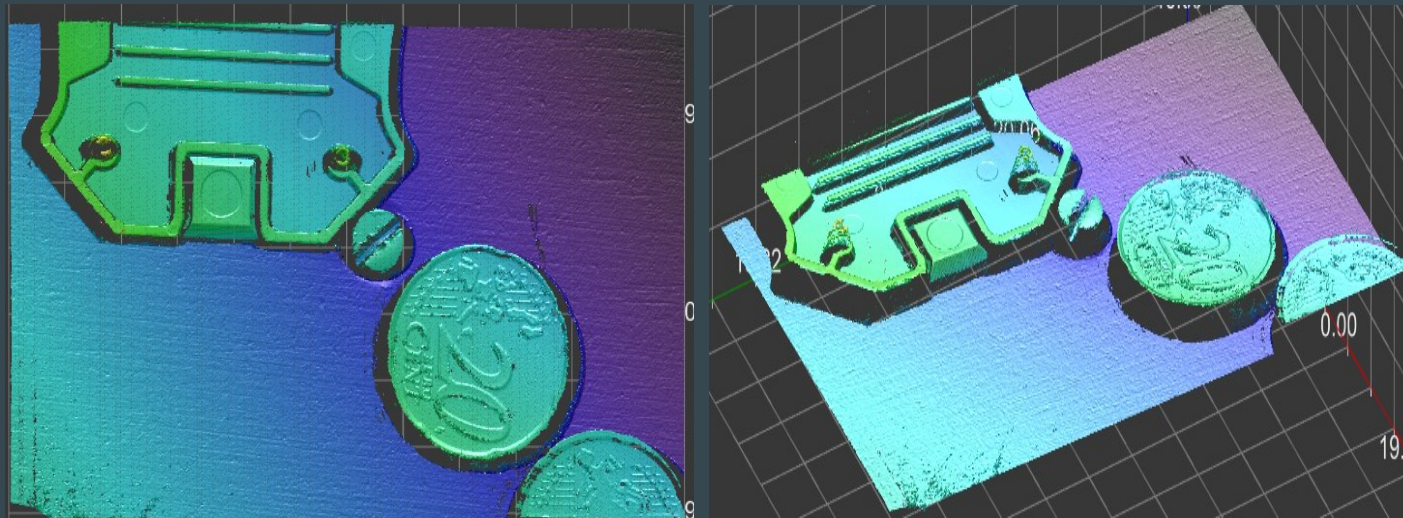
Blue light / 10,000ms Blinking (Scan 10)

Observation



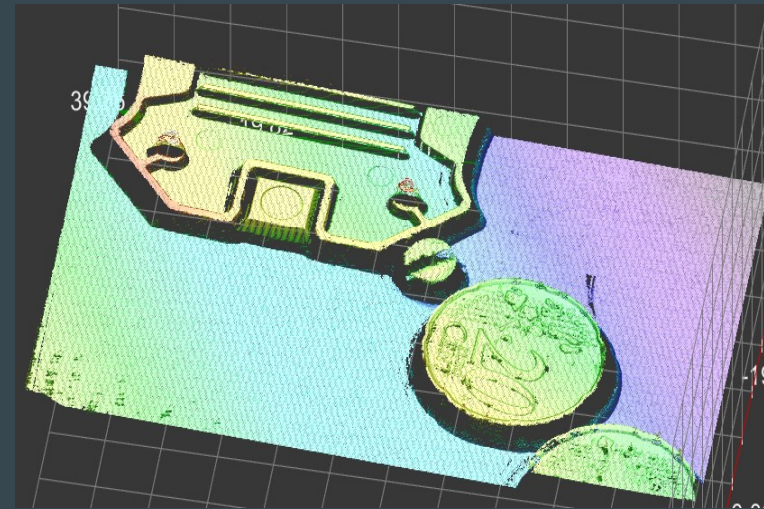
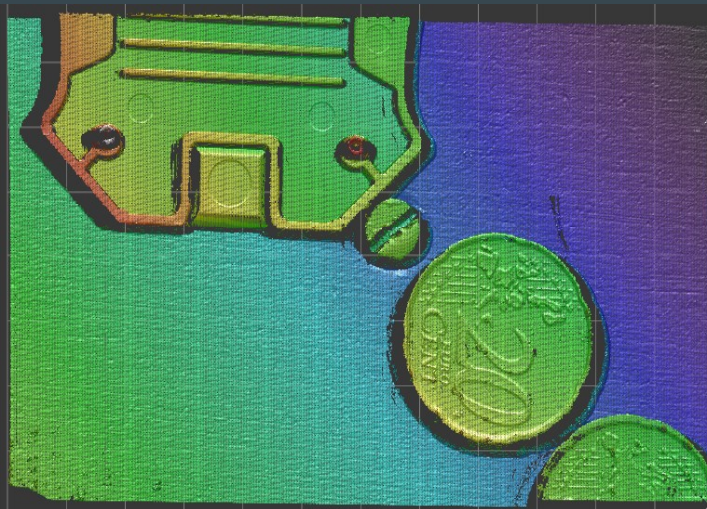
White Light / 30ms Blinking / No room light

Observation



White light / 60ms Blinking

Observation

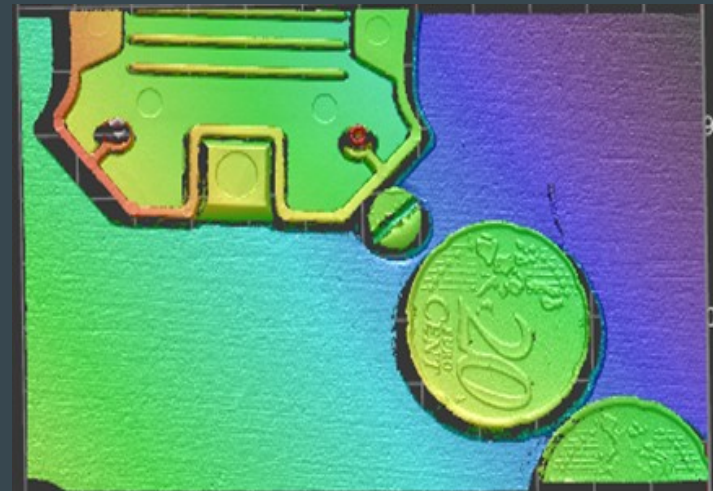


White light / 120ms Blinking

Observation



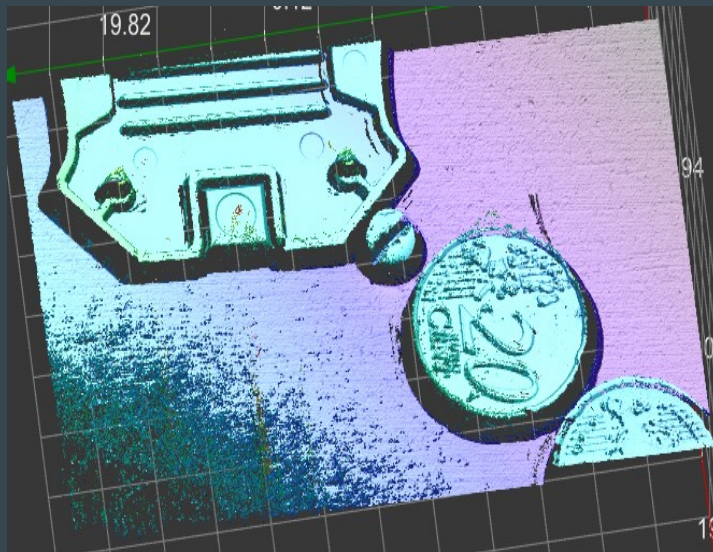
Green light / 30ms Blinking / No room light



Red light / 30ms Blinking / No room light

Observation

50% Brightness of Sensor Lights at 2142x2 Micro seconds exposure, so twice as much as normal, due to brightness half as much.



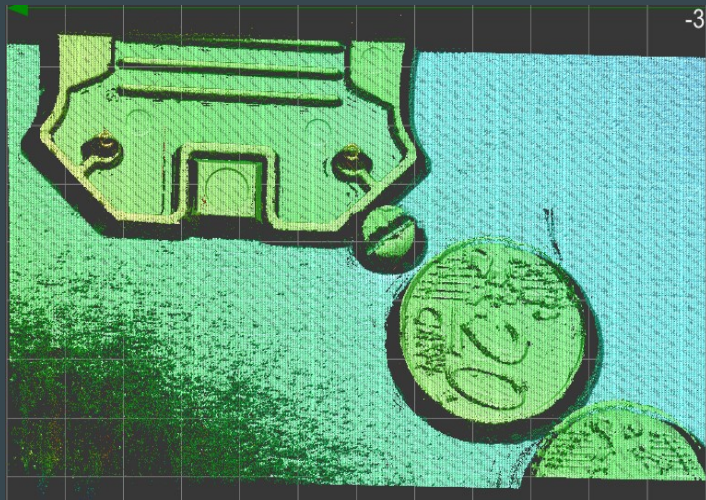
50% White light / 30ms blinking



50% White light / 10ms Blinking

Observation

50% Brightness of Sensor Lights at 2142x2 Micro seconds exposure, so twice as much as normal, due to brightness half as much.



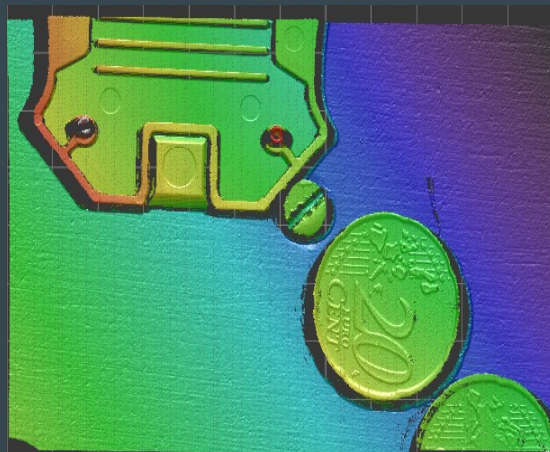
50% White light / 150ms Blinking



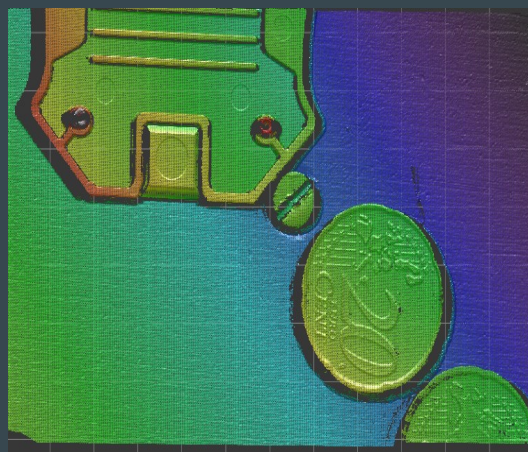
50% White light / 120ms Blinking

Observation

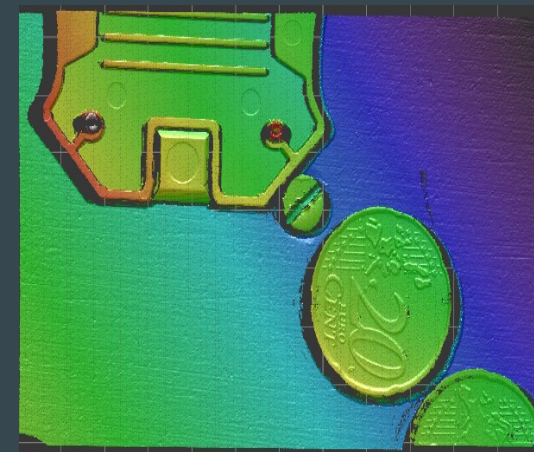
When changing the wavelength, we observe minimal disturbances (Red, Green, Blue) wavelengths /colors or (R,G,B) whereas each value for R, G or B ≤ 255 and sum of R,G,B = 255 (No Blinking effect).



Case 1 (115,0,140)
Equal proportion of R&B



Case 2 (55,0,200)
Dominance of B



Case 3 (195,0,60)
Dominance of R

Analysis

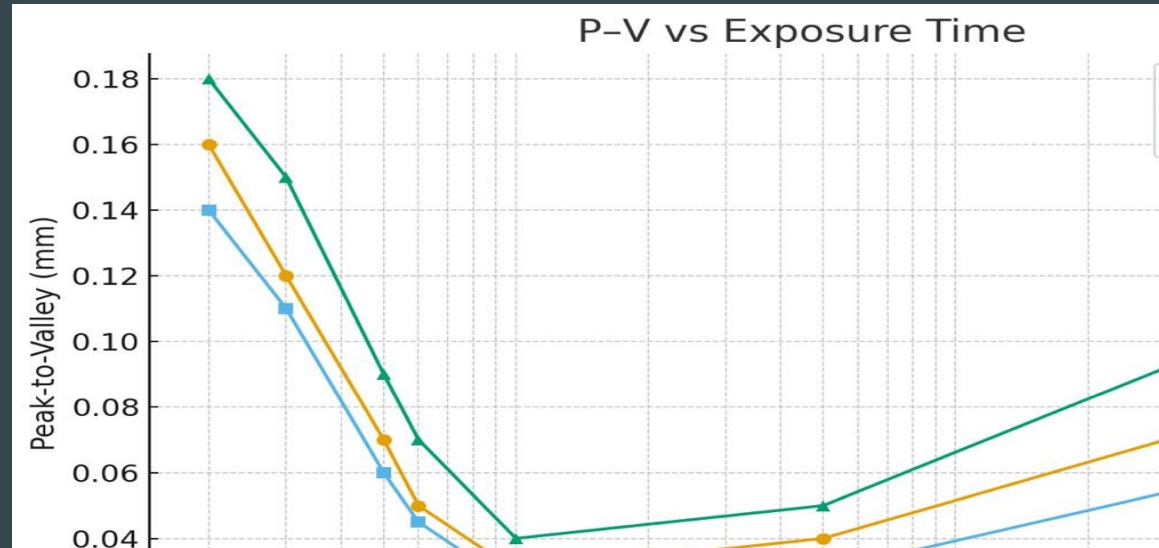
Terms

- **Peak to valley (P-V)**
 - Maximum height difference between the highest point (peak) and the lowest point (valley)
 - Identification of extreme points
- **Sigma: Average variability of the height distribution**
 - General roughness of the surface

Analysis

Data analysis: Peak to Valley (P-V)

Blinking exposure (ms)	P-V Wood	P-V Plastic	P-V Metal
20	0.160	0.140	0.180
30	0.120	0.110	0.150
50	0.070	0.060	0.090
60	0.050	0.045	0.070
100	0.030	0.025	0.040
500	0.040	0.030	0.050
10,000	0.090	0.070	0.120



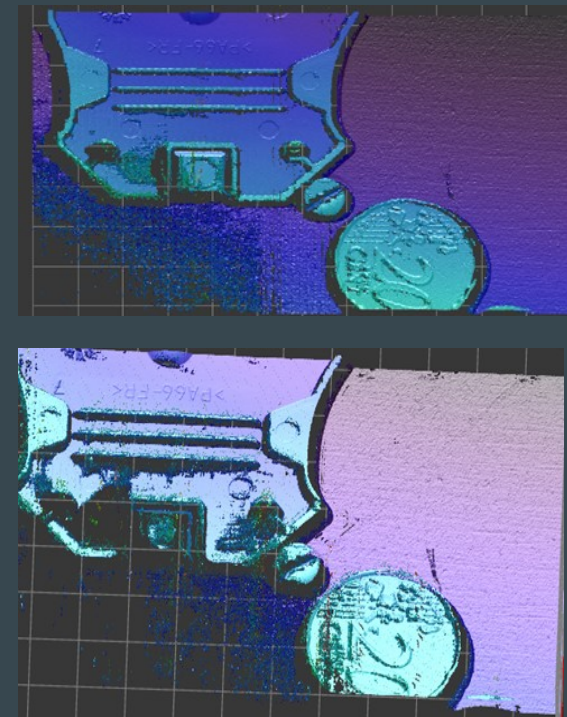
The U-shaped trend confirms exposure optimization: noise dominates at low and high exposure.

- Metal consistently shows the highest P-V (reflectivity errors).
- Plastic gives the smoothest results (diffuse reflection).

Stable, accurate 3D surface at 50-100ms as it reaches minimum.

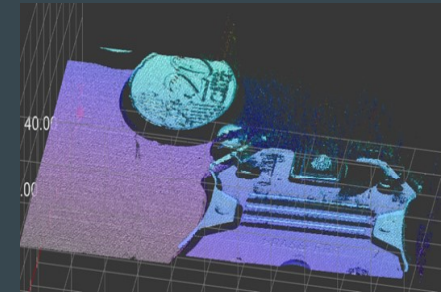
Analysis

- Although the surface appears more disturbed between 20–120 ms due to strong blinking from the ring light and sensor light interference, this disturbance is mainly optical, not geometric.
- The 3D sensor reconstructs height using phase information from the structured pattern - not brightness.
- At short exposures (20–60 ms), the sensor captures random phases of the blinking, producing unstable data and high P-V and σ values.



Analysis

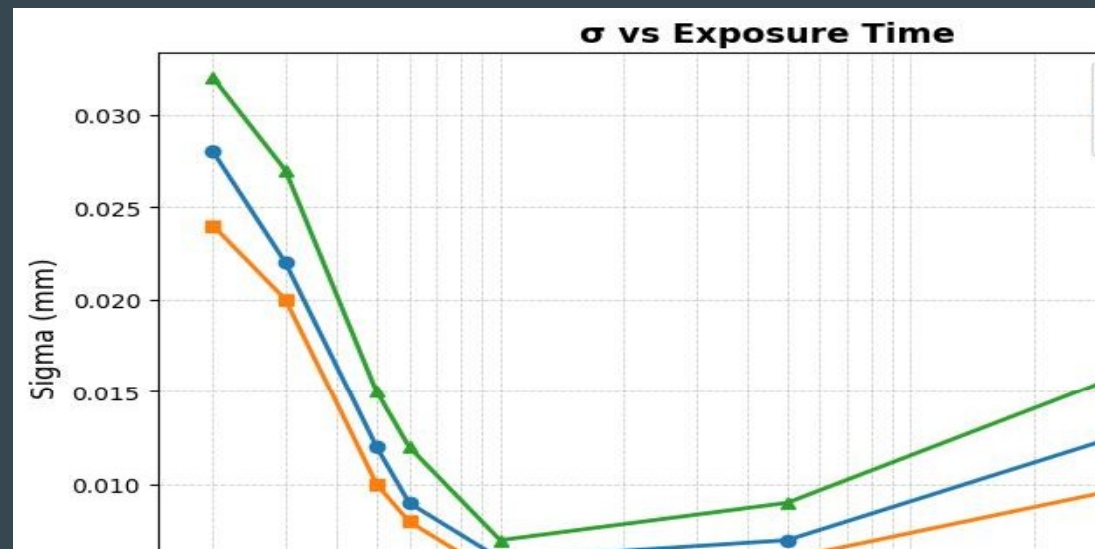
- As exposure increases to 80–120ms, the sensor integrates over multiple blinking cycles, averaging out brightness fluctuations and improving phase stability. This causes P-V and σ to decrease, even though visually the scans seem noisier.
- Visual disturbance $\neq$ Numerical disturbance
- Optical noise, no geometrical noise.
- Signal to noise ratio improves $\rightarrow$ phase stability improves $\rightarrow$ P-V, Sigma drops .
- Longer exposure smooths the blinking effect in phase data, but it looks worse visually, but numerically stable.
- After 500–10,000ms, overexposure and reflections starts saturating pixels and again raise both values. Hence, the trend shows lowest geometric deviation near 100 ms, where the sensor achieves the best phase accuracy despite visible blinking.



Analysis

Data analysis: σ Sigma

Blinking exposure (ms)	σ Wood	σ Plastic	σ Metal
20	0.028	0.024	0.032
30	0.022	0.020	0.027
50	0.012	0.010	0.015
60	0.009	0.008	0.012
100	0.006	0.005	0.007
500	0.007	0.006	0.009
10,000	0.016	0.012	0.020



Sigma decreases from 20 → 100ms → the surface noise reduces as exposure becomes optimal.

- Minimum σ occurs around 100ms (best illumination).
- Rises again beyond 500ms → overexposure and reflection effects.
- Metal > Wood > Plastic throughout - metal reflects more strongly, causing higher variation.

The following behavior was observed during the measurement processes:

Noticeable interference effect:

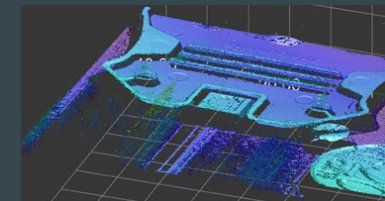
- Multi-reflection/Scattering artefacts (spikes or holes).
- Shiny metal surfaces reflect light specularly, not diffusely.
- Overexposed and dark areas appear where light reflects toward or away from the sensor.
- This leads to false height data.

Cause: The reflected fringe pattern misses the camera, breaking phase information and causing measurement gaps. Reducing exposure or use of Polarisation filters can reduce the effect.



Noticeable interference effect:

- Shadowing and occlusion (Invalid & disappeared 3D data)
 - Region between coin and bracket appears black/dark blue.
 - Line of sight from the sensor is obstructed by nearby geometry.



Applications and Examples in real life situations

- Blinking and disturbances are caused by a **mix of mechanical motion, electrical timing interference, and environmental noise.**
- These manifest visually as **blue/white blink**, or as **depth noise** and **surface distortions** in 3D data.

Summary

- Changing frequency of blue light gives maximum disturbances followed by white light.
- The blinking effect appears stronger with blue structured light because:
 - Blue light has a shorter wavelength and sharper contrast,
 - Sensors and projectors handle blue light differently (less sensitivity, more modulation),
 - Perceptually, both cameras and human vision accentuate blue fluctuations.
- Minimal disturbances when changing the wavelengths of the light or we can say mixing of different colors.
- More the stable light is, produces minimal effects than more blinking light.



*Thank you
for your attention*